

Closed-Form Solution of Markowitz-Mean-Variance Optimization

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Suppose we have a portfolio p consisting of 2 assets A and B , with respective weights w_A and w_B . The Variance of the portfolio p is given by:

$$\text{Var}(p) = \text{Var}(w_A A + w_B B) = \text{Cov}(w_A A + w_B B, w_A A + w_B B) \quad (1)$$

Thanks to the bilinearity of covariance, we can expand the above equation as follows :

$$\begin{aligned} \text{Var}(p) &= w_A^2 \text{Var}(A) + w_B^2 \text{Var}(B) + 2w_A w_B \text{Cov}(A, B) \\ &= w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \sigma_A \sigma_B \rho_{AB}, \text{ where } \rho_{AB} \text{ is the correlation coefficient.} \end{aligned} \quad (2)$$

Thanks to this equation, we can see that indeed 2 assets with negative correlation would reduce the variance and therefore, the volatility squared of our portfolio.

Now, let's consider the case of n assets, with respective weights $\mathbf{w} = (w_1, w_2, \dots, w_n)$ and returns $r_1, r_2, \dots, r_n$. We thus have the return of the portfolio $r_p = \sum_{i=1}^n w_i r_i$. Treating r_i as random variables, we can deduce the following:

$$\text{Var}(r_p) = \text{Cov}(r_p, r_p) = \text{Cov} \left(\sum_{i=1}^n w_i r_i, \sum_{j=1}^n w_j r_j \right) = \sum_{i=1}^n \sum_{j=1}^n w_i w_j \text{Cov}(r_i, r_j) \quad (3)$$

On the other hand, let Σ be a covariance matrix of the returns of the assets, with $\Sigma_{ij} = \text{Cov}(r_i, r_j)$. We therefore have $(\Sigma \mathbf{w})_i = \sum_{j=1}^n \Sigma_{ij} w_j$ and consequently

$$\mathbf{w}^\top \Sigma \mathbf{w} = \sum_{i=1}^n w_i (\Sigma \mathbf{w})_i = \sum_{i=1}^n w_i \sum_{j=1}^n \Sigma_{ij} w_j = \sum_{i=1}^n \sum_{j=1}^n w_i w_j \text{Cov}(r_i, r_j) = \text{Var}(r_p) \quad (4)$$

We can infer directly that reducing the variance of the portfolio, and therefore its volatility, is equivalent to minimizing the quadratic form $\mathbf{w}^\top \Sigma \mathbf{w}$ with respect to the weights $\mathbf{w}$. Assuming we are working with a budget of 1 with a target return of r_p , we can formulate the following optimisation problem:

$$\begin{aligned} \min_{\mathbf{w}} \quad & \mathbf{w}^\top \Sigma \mathbf{w} \\ \text{s.t.} \quad & \mathbf{w}^\top \boldsymbol{\mu} = r_p \\ & \mathbf{w}^\top \mathbf{1} = 1 \end{aligned} \quad (5)$$

We solve this optimization problem via Lagrange multipliers, thus, introducing one multiplier per constraint, our Lagrangian is given by:

$$\begin{aligned} \mathcal{L}(\mathbf{w}, \lambda, \gamma) &= \frac{1}{2} \mathbf{w}^\top \Sigma \mathbf{w} - \lambda (\mathbf{w}^\top \boldsymbol{\mu} - r_p) - \gamma (\mathbf{w}^\top \mathbf{1} - 1) \\ \Rightarrow \frac{\partial \mathcal{L}}{\partial \mathbf{w}} &= \Sigma \mathbf{w} - \lambda \boldsymbol{\mu} - \gamma \mathbf{1} \end{aligned} \quad (6)$$

Setting this partial derivative to zero, we obtain $\mathbf{w}$ as follows:

$$\mathbf{w} = \Sigma^{-1}(\lambda \boldsymbol{\mu} + \gamma \mathbf{1}) \quad (7)$$

Plugging this value of $\mathbf{w}$ into the constraints, we obtain the following system of equations:

$$\lambda \underbrace{\boldsymbol{\mu}^T \Sigma^{-1} \boldsymbol{\mu}}_A + \gamma \underbrace{\boldsymbol{\mu}^T \Sigma^{-1} \mathbf{1}}_B = \bar{r} \quad (8)$$

$$\lambda \underbrace{\mathbf{1}^T \Sigma^{-1} \boldsymbol{\mu}}_B + \gamma \underbrace{\mathbf{1}^T \Sigma^{-1} \mathbf{1}}_C = 1 \quad (9)$$

which can be written as the linear system

$$\begin{pmatrix} A & B \\ B & C \end{pmatrix} \begin{pmatrix} \lambda \\ \gamma \end{pmatrix} = \begin{pmatrix} \bar{r} \\ 1 \end{pmatrix} \quad (10)$$

Denoting $M = \begin{pmatrix} A & B \\ B & C \end{pmatrix}$, we have

$$\begin{aligned} \begin{pmatrix} \lambda \\ \gamma \end{pmatrix} &= M^{-1} \begin{pmatrix} \bar{r} \\ 1 \end{pmatrix} \\ \Rightarrow \begin{pmatrix} \lambda \\ \gamma \end{pmatrix} &= \frac{1}{AC - B^2} \begin{pmatrix} C & -B \\ -B & A \end{pmatrix} \begin{pmatrix} \bar{r} \\ 1 \end{pmatrix} = \frac{1}{AC - B^2} \begin{pmatrix} C\bar{r} - B \\ A - B\bar{r} \end{pmatrix} \\ \Rightarrow \begin{cases} \lambda(\bar{r}) = \frac{C\bar{r} - B}{AC - B^2} \\ \gamma(\bar{r}) = \frac{A - B\bar{r}}{AC - B^2} \end{cases} \end{aligned} \quad (11)$$

Given $\lambda(\bar{r})$ and $\gamma(\bar{r})$ from equation (11), the optimal weights follow directly by substituting back into equation (7):

$$\begin{aligned} \mathbf{w}(\bar{r}) &= \Sigma^{-1}(\lambda(\bar{r}) \boldsymbol{\mu} + \gamma(\bar{r}) \mathbf{1}) \\ \Rightarrow \mathbf{w}(\bar{r}) &= \frac{1}{AC - B^2} \Sigma^{-1}((C\bar{r} - B) \boldsymbol{\mu} + (A - B\bar{r}) \mathbf{1}) \end{aligned} \quad (12)$$

In our implementation, we solved this system of equations using the `numpy.linalg.solve` function, allowing for a more numerically stable computation compared to explicitly inverting the matrix. In order

to trace the curve of the efficient frontier, we simply sweep the target return $\bar{r}$ from the minimum to the maximum expected return of the assets, and compute the corresponding optimal weights $\mathbf{w}(\bar{r})$ and the associated portfolio variance $\text{Var}(r_p) = \mathbf{w}(\bar{r})^\top \Sigma \mathbf{w}(\bar{r})$.